



In this chapter you will learn about...

- ☐ Decimals
- ☐ Multiplication of decimals
- ☐ Division of decimals

DECIMALS

Decimals are fractions whose denominators are 10, 100, 1000, etc.

For example : $\frac{2}{10}$, $\frac{6}{100}$, $\frac{7}{1000}$, etc. are all decimal fractions and can be written as 0.2, 0.06 and 0.007 respectively.

The numbers written in decimal form are called **decimal numbers**. The number of digits contained in the decimal part of a decimal number is known as number of **decimal places**.

Example : 3.456 has 3 decimal places. Let us review what we have learnt in class VI and then take up multiplication and division of decimals.

Like and Unlike Decimals

Decimal fractions having same number of decimal places are known as **like decimals**.

Examples : 5.17, 19.25, 216.42 are all like decimals.

Decimal fractions having different number of decimal places are known as **unlike decimals**.

Examples : 12.5, 13.72, 18.564 are all unlike decimals.

Comparison of Decimals

To compare decimal fractions we may follow the following steps :

- Step 1 : Convert the given decimals into like decimals.
 - Step 2 : Compare whole number (integral) parts. The decimal number with greater integral part will be greater.
 - Step 3 : If integral parts are equal, compare the digits at tenths place. The decimal number with the greater digit in the tenths place will be greater.
 - Step 4 : If the tenths digits are same, proceed for hundredths place digits. The decimal number having greater digit at hundredths place will be greater.
- Similarly, proceed further.

Example 1 : Which is greater : 15.4 or 15.36?

Solution : Given decimals are unlike. The like decimals will be 15.40 and 15.36.

As the integral part of both decimals is same, so comparing the tenths place, we get, $15.4 > 15.36$

Example 2 : Write the following decimals in ascending order :

2.53, 2.0532, 4.505, 4.0503, 4.3502

Solution : Converting the given decimals into like decimals, we get them as

2.5300, 2.0532, 4.5050, 4.0503, 4.3502.

Clearly,

$2.0532 < 2.5300 < 4.0503 < 4.3502 < 4.5050$

Hence, the given decimals in ascending order are

2.0532, 2.53, 4.0503, 4.3502, 4.505.

Decimals as Fractions

We convert the given decimal into an equivalent fraction with denominator 10, 100 or 1000 depending on the places of decimal after removing the decimal point from the numerator.

Example 3 : Express each of the following as a fraction :

(i) 0.05

(ii) 0.75

(iii) 2.064

(iv) 2.75

Solution : (i) $0.05 = \frac{5}{100} = \frac{1}{20}$

(ii) $0.75 = \frac{75}{100} = \frac{3}{4}$

(iii) $2.064 = \frac{2064}{1000} = \frac{258}{125}$

(iv) $2.75 = \frac{275}{100} = \frac{11}{4}$

Fractions as Decimals

Step 1 : Convert the given fraction into an equivalent fraction with denominator 10, 100 or 1000, etc.

Step 2 : Keep the decimal point before the number in the numerator as many digits as number of zeros in the denominator.

Step 3 : If the number of digits in the numerator is not equal to the number of zeros in the denominator, then add zeros to the left of the number to make it equal.

Step 4 : In case of mixed fractions, keep the whole number as the integral part and proceed as in step 1, 2 and 3.

Example 4 : Express the following fractions as decimals :

(i) $\frac{6}{10}$

(ii) $\frac{8}{100}$

(iii) $7\frac{1}{4}$

(iv) $12\frac{1}{8}$

Solution : (i) $\frac{6}{10} = 0.6$

(Inserting the decimal point one place to the left)

(ii) $\frac{8}{100} = 0.08$

(Inserting the decimal point two places to the left)

(iii) $7\frac{1}{4} = 7 + \frac{1}{4} = 7 + \frac{25}{4 \times 25}$

$= 7 + \frac{25}{100} = 7 + 0.25 = 7.25$

(Inserting the decimal point two places to the left)

$$(iv) 12\frac{1}{8} = 12 + \frac{1}{8}$$

$$= 12 + \frac{1 \times 125}{8 \times 125} = 12 + \frac{125}{1000}$$

$$= 12 + 0.125 = 12.125 \text{ (Inserting the decimal point three places to the left)}$$

Addition and Subtraction of Decimals

In order to add or subtract decimals, we may use the following steps :

Step 1 : Convert the given decimals into like decimals.

Step 2 : Arrange the digits as per place value in columns with their decimal point directly below each other.

Step 3 : Add or subtract as we do in case of numerals.

Step 4 : Decimal point is placed in the same place as the numbers above it.

Example 5 : Add the following decimals :

$$(i) 8.6 + 15.123 \quad (ii) 0.33 + 33.014 \quad (iii) 5.46 + 2.374 + 12.8$$

Solution : (i) Converting the given decimals into like decimals, we have

$$8.6 + 15.123 = 8.600 + 15.123$$

$$= 23.723$$

$$\begin{array}{r} 8.600 \\ + 15.123 \\ \hline 23.723 \end{array}$$

(ii) Converting the given decimals into like decimals, we have

$$0.33 + 33.014 = 0.330 + 33.014$$

$$= 33.344$$

$$\begin{array}{r} 0.330 \\ + 33.014 \\ \hline 33.344 \end{array}$$

(iii) Converting the given decimals into like decimals, we have

$$5.46 + 2.374 + 12.8 = 5.460 + 2.374 + 12.800$$

$$= 20.634$$

$$\begin{array}{r} 5.460 \\ + 2.374 \\ + 12.800 \\ \hline 20.634 \end{array}$$

Example 6 : Subtract :

$$(i) 3.72 \text{ from } 8.47$$

$$(iii) 27.4 \text{ from } 40.056$$

$$(ii) 16.92 \text{ from } 24.08$$

$$(iv) 6.354 \text{ from } 10$$

Solution :

$$(i) \quad 8.47$$

$$\begin{array}{r} 8.47 \\ - 3.72 \\ \hline 4.75 \end{array}$$

$$(ii) \quad 24.08$$

$$\begin{array}{r} 24.08 \\ - 16.92 \\ \hline 7.16 \end{array}$$

$$(iii) \quad 40.056$$

$$\begin{array}{r} 40.056 \\ - 27.400 \\ \hline 12.656 \end{array}$$

$$(iv) \quad 10.000$$

$$\begin{array}{r} 10.000 \\ - 6.354 \\ \hline 3.646 \end{array}$$

Remember : Adding zeros after the last digit of the decimal part of any decimal number does not change the value of the decimal.



EXERCISE 3.1

- Convert each of the following decimals into fractions in lowest form :
(i) 0.8 (ii) 2.25 (iii) 17.5 (iv) 0.0375
(v) 0.524
- Write each of the following as decimals :
(i) $\frac{29}{10}$ (ii) $15\frac{1}{8}$
(iii) $234\frac{1}{25}$ (iv) $18 + \frac{3}{10} + \frac{4}{1000}$
- Add the following :
(i) 4.56, 0.8 and 32.5 (ii) 0.05 and 0.004
(iii) 13.225 and 9.25 (iv) 14.44, 6.234 and 18
- Find the value of :
(i) $8.752 - 4.9$ (ii) $23.8 - 0.25$
(iii) $100 - 36.48$ (iv) $8.5 - 2.785$
- Simplify :
(i) $68.5 - 21.45 + 26.2 - 13.46$ (ii) $6.6 + 23.8 - 0.99$
(iii) $34.26 - 15.029 + 3.04$ (iv) $6 - 2.5307 + 3.1$
- Arrange the following decimals in ascending order :
(i) 4.6, 0.22, 24.13, 20.05 (ii) 0.52, 36.28, 2.05, 3.5

MULTIPLICATION OF DECIMALS

Multiplication of a Decimal by 10, 100, 1000, etc.

- Rules :**
- On multiplying a decimal by 10, the decimal point is shifted to the right by one place.
 - On multiplying a decimal by 100, the decimal point is shifted to the right by two places.
 - On multiplying a decimal by 1000, the decimal point is shifted to the right by three places.

Example 1 : Find the product :

- (i) 7.24×10 (ii) 7.24×100 (iii) 7.24×1000
(iv) 25.46×10 (v) 127.354×10

Solution : (i) 7.24×10

As there is one zero in multiplier (10), therefore, decimal point will shift by one place to the right while multiplying.

$$7.24 \times 10 = 72.4$$

(ii) $7.24 \times 100 = 724$ (decimal point shifts two places to the right)

(iii) $7.24 \times 1000 = 7.240 \times 1000 = 7240$
(decimal point shifts three places to the right)

(iv) $25.46 \times 10 = 254.6$ (decimal point shifts one place to the right)

(v) $127.354 \times 10 = 1273.54$ (decimal point shifts one place to the right)

Multiplication of a Decimal by a Whole Number

To multiply a decimal by a whole number, we may follow the following steps :

Step 1 : Multiply the given decimal without the decimal point by the given whole number.

Step 2 : Mark the decimal point in the product to have as many places of decimal as there are in the given decimal.

Example 2 : Multiply the following :

(i) 4.25×12 (ii) 0.203×16 (iii) 0.67×15 (iv) 0.0361×18

(v) 0.0008×9

Solution : (i) 4.25×12

$$425 \times 12 = 5100$$

(First multiplying without decimal point)

$$4.25 \times 12 = 51.00$$

(Put the decimal point after two digits)

$$\therefore 4.25 \times 12 = 51$$

(ii) 0.203×16

$$203 \times 16 = 3248$$

(First multiplying without decimal point)

$$\therefore 0.203 \times 16 = 3.248$$

(Put the decimal point after three digits)

(iii) 0.67×15

$$67 \times 15 = 1005$$

(First multiplying without decimal point)

$$\therefore 0.67 \times 15 = 10.05$$

(Put the decimal point after two digits)

(iv) 0.0361×18

$$361 \times 18 = 6498$$

(First multiplying without decimal point)

$$\therefore 0.0361 \times 18 = 0.6498$$

(Put the decimal point after four digits)

(v) 0.0008×9

$$8 \times 9 = 72$$

(First multiplying without decimal point)

$$\therefore 0.0008 \times 9 = 0.0072$$

(Put the decimal point after four digits)

Multiplication of a Decimal by a Decimal

We may formulate the following method for multiplying a decimal by a decimal.

Step 1 : Multiply the two decimals without the decimal point.

Step 2 : Put the decimal point in the product in such a way that the number of decimal places in the product is equal to the sum of the number of decimal places in the given decimals.

Example 3 : Multiply : 62.48×5.9

Solution : 62.48×5.9

First, we multiply two decimals without the decimal point.

$$6248 \times 59$$

$$6248$$

$$\times 59$$

$$\hline 56232$$

$$+ 312400$$

$$\hline 368632$$

$$\begin{array}{ccc} 62.48 \times 5.9 & = & 368.632 \\ \text{2 decimal} & \text{1 decimal} & \text{2 + 1 = 3 decimal} \\ \text{places} & \text{place} & \text{places} \end{array}$$

Example 4 : Multiply : 0.032×0.416

Solution : 0.032×0.416

We have, $32 \times 416 = 13312$

$$\begin{array}{r} 416 \\ \times 32 \\ \hline 832 \\ + 12480 \\ \hline 13312 \end{array}$$

$$\begin{array}{ccc} 0.032 & \times & 0.416 \\ \text{3 decimal} & & \text{3 decimal} \\ \text{places} & & \text{places} \end{array} = \begin{array}{c} 0.013312 \\ \text{3+3=6 decimal} \\ \text{places} \end{array}$$

Hence, $0.032 \times 0.416 = 0.013312$

Example 5 : Multiply 0.0243 by 0.0526.

Solution : First, we multiply 243 by 526.

$$\begin{array}{r} 243 \\ \times 526 \\ \hline 1458 \\ + 4860 \\ + 121500 \\ \hline 127818 \end{array}$$

$$\therefore 243 \times 526 = 127818$$

$$\begin{array}{ccc} 0.0243 & \times & 0.0526 \\ \text{4 decimal} & & \text{4 decimal} \\ \text{places} & & \text{places} \end{array} = \begin{array}{c} 0.00127818 \\ \text{4+4=8} \\ \text{decimal places} \end{array}$$

Hence, $0.0243 \times 0.0526 = 0.00127818$

Example 6 : Find the product : $0.457 \times 26.8 \times 23$.

Solution : First, we find the product $457 \times 268 \times 23$.

$$457 \times 268 \times 23 = 457 \times (268 \times 23)$$

$$\begin{array}{r} 268 \\ \times 23 \\ \hline 804 \\ + 5360 \\ \hline 6164 \end{array}$$

$$\therefore 457 \times (268 \times 23) = 457 \times 6164$$

$$\begin{array}{r} 6164 \\ \times 457 \\ \hline 43148 \\ + 308200 \\ + 2465600 \\ \hline 2816948 \end{array}$$

Therefore, $457 \times 268 \times 23 = 2816948$

$$\begin{array}{ccccccc} 0.457 & \times & 26.8 & \times & 23 & = & 281.6948 \\ \text{3 decimal places} & & \text{1 decimal place} & & & & \text{(3 + 1 = 4 decimal places)} \end{array}$$

Hence, $0.457 \times 26.8 \times 23 = 281.6948$



EXERCISE 3.2

1. Evaluate :

(i) 2.6×10

(iv) 234.05×10

(vii) 3.57×100

(x) 0.08×100

(ii) 34.7×10

(v) 21.1×10

(viii) 48.06×100

(xi) 0.09×1000

(iii) 128.6×10

(vi) 352.4×100

(ix) 0.7×10

(xii) 0.06×1000

2. Evaluate :

(i) 0.6×6

(iv) 25.2×7

(vii) 3×0.55

(ii) 6.23×5

(v) 0.08×5

(iii) 9×2.4

(vi) 351.06×4

3. Evaluate :

(i) 6.2×0.2

(iv) 0.3×0.03

(vii) 14.3×0.12

(x) 101.01×0.01

(ii) 21.7×0.1

(v) 0.23×19.4

(viii) 1.05×0.03

(xi) 200.02×2.2

(iii) 0.4×302.5

(vi) 20.05×2.05

(ix) 30.02×2.02

4. Find the area of a square field of side 5.8 cm.

5. 1 kg of sugar costs ₹ 25.30. Find the cost of 13.5 kg of sugar.

6. If 1 metre of cloth costs ₹ 158.50, what is the cost of 4.5 m of cloth?

7. If the cost of a book is ₹ 29.75, find the cost of 32 such books.

8. Find the area of a rectangle whose length is 18.60 m and breadth is 7.05 m.

9. Find the product :

(i) $1.2 \times 1.2 \times 0.012$

(iii) $33.3 \times 3.3 \times 0.33$

(ii) $3.2 \times 1.5 \times 2.5$

(iv) $0.6 \times 0.02 \times 0.003$

DIVISION OF DECIMALS

A vehicle covers a distance of 46.5 km in 2.5 litres of petrol. Find the distance covered in 1 litre of petrol.

$$\text{Distance covered in 1 litre of petrol} = \frac{46.5}{2.5} \text{ km} = 18.6 \text{ km}$$

In our daily life, we come across various situations where we need to know the division of decimals.

Division of a Decimal by 10, 100 and 1000

Consider $23.5 \div 10 = \frac{235}{10} \times \frac{1}{10} = \frac{235}{100} = 2.35$

Similarly, $23.5 \div 100 = \frac{235}{10} \times \frac{1}{100} = \frac{235}{1000} = 0.235$

Also, $23.5 \div 1000 = \frac{235}{10} \times \frac{1}{1000} = \frac{235}{10000} = 0.0235$

Thus, while dividing a decimal number by 10, 100 or 1000, the digits of the number and the quotient are same but the decimal point in the quotient shifts to the left by as many places as there are zeros in the denominator.

Hence, $3.25 \div 10 = 0.325$; $3.25 \div 100 = 0.0325$ and $3.25 \div 1000 = 0.00325$.

Division of a Decimal Number by a Whole Number

In order to divide a decimal number by a whole number, we may formulate the following method :

Step 1 : Perform the division by considering the dividend a whole number.

Step 2 : When the division of whole-number part of the dividend is complete, put the decimal point in the quotient and proceed with the division.

Step 3 : If the last remainder is non-zero, insert as many zeros on the right of decimal part of the dividend as needed.

The following examples will illustrate the procedure.

Example 1 : Divide : (i) 62.64 by 12

$$\begin{array}{r} 5.22 \\ 12 \overline{) 62.64} \\ \underline{-60} \\ 26 \\ \underline{-24} \\ 24 \\ \underline{-24} \\ 0 \end{array}$$

$$\therefore 62.64 \div 12 = 5.22$$

(ii) 0.672 by 14

$$\begin{array}{r} 0.048 \\ 14 \overline{) 0.672} \\ \underline{-0} \\ 67 \\ \underline{-56} \\ 112 \\ \underline{-112} \\ 0 \end{array}$$

$$\therefore 0.672 \div 14 = 0.048$$

Example 2 : Divide : (i) 9.27 by 15

$$\begin{array}{r} 0.618 \\ 15 \overline{) 9.270} \\ \underline{-0} \\ 92 \\ \underline{-90} \\ 27 \\ \underline{-15} \\ 120 \\ \underline{-120} \\ 0 \end{array}$$

$$\therefore 9.27 \div 15 = 0.618$$

(ii) 113.44 by 8

$$\begin{array}{r} 14.18 \\ 8 \overline{) 113.44} \\ \underline{-8} \\ 33 \\ \underline{-32} \\ 14 \\ \underline{-8} \\ 64 \\ \underline{-64} \\ 0 \end{array}$$

$$\therefore 113.44 \div 8 = 14.18$$

Example 3 : Divide : (i) 0.5306 by 5

Solution : (i)
$$\begin{array}{r} 0.10612 \\ 5 \overline{) 0.53060} \\ \underline{-0} \\ 5 \\ \underline{-5} \\ 30 \\ \underline{-30} \\ 6 \\ \underline{-5} \\ 10 \\ \underline{-10} \\ 0 \end{array}$$

$$\therefore 0.5306 \div 5 = 0.10612$$

(ii) 0.7682 by 16

(ii)
$$\begin{array}{r} 0.0480125 \\ 16 \overline{) 0.7682000} \\ \underline{-0} \\ 76 \\ \underline{-64} \\ 128 \\ \underline{-128} \\ 20 \\ \underline{-16} \\ 40 \\ \underline{-32} \\ 80 \\ \underline{-80} \\ 0 \end{array}$$

$$\therefore 0.7682 \div 16 = 0.0480125$$

Division of a Decimal Number by Another Decimal Number

The steps followed for division are as follows :

Step 1 : Multiply the divisor and dividend by 10, 100 or 1000 etc. so that the divisor becomes a whole number.

Step 2 : Then use the same steps for division of a decimal number by a whole number.

Let us take some examples to understand it further.

Example 1 : Divide :

(i) 30.94 by 0.7

(ii) 76.5 by 0.15

(iii) 2.73 by 1.3

(iv) 37.8 by 1.4

Solution : (i)
$$\frac{30.94}{0.7} = \frac{30.94 \times 10}{0.7 \times 10} = \frac{309.4}{7}$$

$$\begin{array}{r} 44.2 \\ 7 \overline{) 309.4} \\ \underline{-28} \\ 29 \\ \underline{-28} \\ 14 \\ \underline{-14} \\ 0 \end{array}$$

$$\therefore 30.94 \div 0.7 = 44.2$$

(ii)
$$\frac{76.5}{0.15} = \frac{76.5 \times 100}{0.15 \times 100} = \frac{7650}{15}$$

$$\begin{array}{r} 510 \\ 15 \overline{) 7650} \\ \underline{-75} \\ 15 \\ \underline{-15} \\ 0 \end{array}$$

$$\therefore 76.5 \div 0.15 = 510$$

(iii)
$$\frac{2.73}{1.3} = \frac{2.73 \times 10}{1.3 \times 10} = \frac{27.3}{13}$$

$$\begin{array}{r} 2.1 \\ 13 \overline{) 27.3} \\ \underline{-26} \\ 13 \\ \underline{-13} \\ 0 \end{array}$$

$$\therefore 2.73 \div 1.3 = 2.1$$

(iv)
$$\frac{37.8}{1.4} = \frac{37.8 \times 10}{1.4 \times 10} = \frac{378}{14}$$

$$\begin{array}{r} 27 \\ 14 \overline{) 378} \\ \underline{-28} \\ 98 \\ \underline{-98} \\ 0 \end{array}$$

$$\therefore 37.8 \div 1.4 = 27$$

Example 2 : Divide : (i) 1.001 by 14

(ii) 0.768 by 16

Solution : (i)
$$\begin{array}{r} 0.0715 \\ 14 \overline{) 1.0010} \\ \underline{-0} \\ 100 \\ \underline{-98} \\ 21 \\ \underline{-14} \\ 70 \\ \underline{-70} \\ 0 \end{array}$$

$\therefore 1.001 \div 14 = 0.0715$

(ii)
$$\begin{array}{r} 0.048 \\ 16 \overline{) 0.768} \\ \underline{-0} \\ 76 \\ \underline{-64} \\ 128 \\ \underline{-128} \\ 0 \end{array}$$

$\therefore 0.768 \div 16 = 0.048$

Example 3 : Divide 26.313 by 1.75.

Solution :
$$26.313 \div 1.75 = \frac{26.313}{1.75} = \frac{26.313 \times 100}{1.75 \times 100} = \frac{2631.3}{175}$$

$$\begin{array}{r} 15.036 \\ 175 \overline{) 2631.30} \\ \underline{-175} \\ 881 \\ \underline{-875} \\ 630 \\ \underline{-525} \\ 1050 \\ \underline{-1050} \\ 0 \end{array}$$

Hence, $26.313 \div 1.75 = 15.036$

Example 4 : Divide 3.411 by 37.5.

Solution :
$$3.411 \div 37.5 = \frac{3.411}{37.5} = \frac{3.411 \times 10}{37.5 \times 10} = \frac{34.11}{375}$$

$$\begin{array}{r} 0.09096 \\ 375 \overline{) 34.11000} \\ \underline{-0} \\ 3411 \\ \underline{-3375} \\ 3600 \\ \underline{-3375} \\ 2250 \\ \underline{-2250} \\ 0 \end{array}$$

Hence, $3.411 \div 37.5 = 0.09096$

Example 5 : The area of a rectangle having breadth 2.6 m is 19.50 sq.m. Find the length of the rectangle.

Solution : Area of rectangle = 19.50 sq.m
Breadth of rectangle = 2.6 m

VALUE CORNER

A taxi driver charges ₹ 27.20 per km. Rahul hired the taxi for a day and travelled 85.4 km. The taxi driver charged him ₹ 2500.

- How much did Rahul pay extra?
- Do you think the taxi driver was honest? Why/why not?

$$\begin{aligned}\text{Length of rectangle} &= \frac{\text{Area}}{\text{Breadth}} \\ &= \frac{19.50}{2.6} = \frac{19.50 \times 10}{2.6 \times 10} = \frac{195}{26} \\ &= 7.5 \text{ m}\end{aligned}$$

$$\begin{array}{r} 7.5 \\ 26 \overline{) 195.0} \\ \underline{-182} \\ 130 \\ \underline{-130} \\ 0 \end{array}$$

Hence, length of the rectangle is 7.5 m.

Example 6 : The product of two decimals is 3.392. If one of them is 1.6, find the other.

Solution : Product of two decimals = 3.392

One decimal number = 1.6

$$\begin{aligned}\text{Other decimal number} &= \frac{\text{Product of two decimals}}{\text{One decimal number}} \\ &= \frac{3.392}{1.6} \\ &= \frac{3.392 \times 10}{1.6 \times 10} \\ &= \frac{33.92}{16} = 2.12\end{aligned}$$

$$\begin{array}{r} 2.12 \\ 16 \overline{) 33.92} \\ \underline{-32} \\ 19 \\ \underline{-16} \\ 32 \\ \underline{-32} \\ 0 \end{array}$$

Hence, the other decimal number is 2.12.



EXERCISE 3.3

1. Divide :

- (i) 2.5 by 10
- (iv) 55.6 by 10
- (vii) 0.56 by 10
- (x) 233.7 by 100

- (ii) 2.5 by 100
- (v) 5.56 by 100
- (viii) 3.26 by 10

- (iii) 2.5 by 1000
- (vi) 3.69 by 10
- (ix) 3.79 by 100

2. Divide :

- (i) 25.5 by 0.5
- (iv) 15.2 by 0.8
- (vii) 89.1 by 3.3

- (ii) 22.5 by 2.5
- (v) 27 by 0.03
- (viii) 2.73 by 1.3

- (iii) 20.4 by 6
- (vi) 12.5 by 2.5
- (ix) 0.5 by 0.05

3. Divide :

- (i) 12.98 by 4
- (iv) 7.75 by 25
- (vii) 24.66 by 12

- (ii) 36 by 0.4
- (v) 11.2 by 16
- (viii) 6.2 by 248

- (iii) 30.94 by 7
- (vi) 1.07 by 2

4. Convert into decimal number :

- (i) 12 + 8
- (iv) 796 + 3184
- (vii) 17 + 40

- (ii) 24 + 25
- (v) 3 + 8
- (viii) 3 + 25

- (iii) 135 + 2
- (vi) 6 + 125
- (ix) 13 + 125

5. The cost of 15 notebooks is ₹ 144. Find the cost of one such notebook.
6. The total weight of certain bags of rice is 650.16 kg. If each bag weighs 10.32 kg, find the number of bags.
7. The product of two decimals is 32.396. If one of the decimals is 5.2, find the other.
8. The cost of one dozen of apples is ₹ 256.20. Find the cost of one apple.
9. The cost of 26 toys is ₹ 1107.60. Find the cost of one such toy.
10. The cost of 10.85 m of ribbon is ₹ 238.70. Find the cost of one metre of ribbon.



FACTS TO REMEMBER

- The fractions in which the denominators are 10, 100, 1000, etc. are known as decimal fractions.
- A decimal has two parts—the whole number part and the decimal part.
- Decimals having the same number of decimal places are called like decimals, otherwise they are known as unlike decimals.
- By adding zeros after the last digit of the decimal part of any decimal number does not change the value of the decimal number e.g., $0.1 = 0.10 = 0.100$, $0.2 = 0.20 = 0.200$.
- Unlike decimals can be converted into like decimals by adding requisite number of zeros at the end of the decimal part.
- While multiplying two decimal numbers, first multiply them as whole numbers. Count the number of digits to the right of the decimal point in both the decimal numbers. Add the number of digits counted. Put the decimal point in the product by counting the digits from its rightmost place.
For example, $0.3 \times 0.7 = 0.21$.
- To multiply the decimal number by 10, 100 or 1000, the decimal point is shifted to the right by one, two or three places respectively.
- To divide a decimal number by a whole number, we first divide them as whole numbers. Then, we place the decimal point in the quotient as in the decimal number.
- To divide a decimal number by 10, 100 or 1000, the decimal point is shifted to the left by one, two or three places respectively.
- While dividing two decimal numbers, convert the divisor into a whole number by multiplying the dividend and the divisor by 10, 100 or 1000 etc. and then divide the new dividend by the whole number obtained above.



MULTIPLE CHOICE QUESTIONS

Choose the correct alternatives in each of the following :

1. In a division the quotient is 0.48, divisor is 5 and remainder is zero, then dividend is
(a) 2.04 (b) 4.2 (c) 2.40 (d) 4.02
2. $\frac{1}{5}$ can be expressed in decimal as
(a) 0.5 (b) 0.1 (c) 0.02 (d) 0.2
3. 0.60 can be written as
(a) $\frac{6}{10}$ (b) $\frac{6}{100}$ (c) $\frac{6}{1000}$ (d) none of these
4. How many $\frac{1}{100}$ together make 1?
(a) 10 (b) 1 (c) 100 (d) none of these
5. The decimal representation of $4 + \frac{5}{10} + \frac{7}{100}$ is
(a) 0.457 (b) 0.57 (c) 45.7 (d) 4.57
6. The lowest form of the decimal 0.004 is
(a) $\frac{1}{20}$ (b) $\frac{1}{250}$ (c) $\frac{1}{150}$ (d) $\frac{1}{25}$
7. 0.24 when expressed in the form $\frac{p}{q}$ is
(a) $\frac{6}{25}$ (b) $\frac{4}{25}$ (c) $\frac{24}{10}$ (d) $\frac{3}{25}$
8. 3700 grams =
(a) 0.37 kg (b) 3.7 kg (c) 37 kg (d) 0.037 kg
9. Which of the following is a true statement?
(a) $1.16 > 1.4$ (b) $1.16 < 1.2$ (c) $1.163 > 1.170$ (d) $1.14 < 1.040$
10. $1.004 - 0.4$ is equal to
(a) 0.006 (b) 1 (c) 0.604 (d) 0.640



MENTAL MATHS CORNER

Fill in the blanks :

1. If $7.645 \times 4.8 = 36.696$, then $76.45 \times 0.48 =$
2. $31.4965 \times$ $= 3149.65$.
3. $2.89 + 1.7$ is equal to
4. $\frac{1}{200}$ in decimal can be written as
5. 25 when multiplied by gives the product 6.25.
6. $342 \text{ cm} =$ metre.
7. $(4.26 - 3.26) \div 100 =$
8. On multiplying a decimal by 100, the decimal point is shifted to the by two places.
9. If one bag of sugar weighs 8 kg 500 g, then 10 bags will weigh
10. The product of two decimals is 1.56. If one of them is 1.2, then other is



REVIEW EXERCISE

1. Arrange the following decimals in ascending order :
7.46, 7.64, 7.6, 7.4, 7.06, 7.04
2. Convert each of the following into decimals :
(i) $\frac{4}{5}$ (ii) $\frac{6528}{1000}$ (iii) $\frac{39}{25}$ (iv) $3\frac{5}{8}$
3. Add : 19.8, 7.26, 0.074 and 2.37.
4. Subtract 3.6204 from 7. .
5. Find the product :
(i) 32.5×1000 (ii) 0.237×15 (iii) 0.0065×4 (iv) 0.327×12
6. Find the product :
(i) 7.4×2.6 (ii) 4.26×0.08 (iii) 0.016×0.26 (iv) 0.004×0.39
7. Divide :
(i) 0.068 by 0.004 (ii) 217.35 by 6.3 (iii) 7804.5 by 104.06 (iv) 0.06764 by 0.089
8. The product of two decimals is 0.768. If one of the decimal numbers is 1.6, find the other.
9. A taxi driver charges ₹ 687.30 for a journey of 94.8 km. What will be the charges for a journey of 10 km?

- The side of a square is 4.05 metre, find its area.
- Find the area of a rectangle whose sides are 13.25 m and 5.10 m.
- The cost of a flower vase is ₹ 32.75. Find the cost of 11 such flower vases.
- Rashmi cuts 90 m of cloth into pieces of 1.20 m each. How many pieces does she get?



HOTS QUESTIONS

- Each side of a regular polygon is 3.2 cm in length. The perimeter of the polygon is 19.2 cm. How many sides does the polygon have? What is the name of the polygon?
- A basket has 2 pineapples for ₹ 13.25 each, 5 apples for ₹ 4.50 each, 4 bananas for ₹ 2.75 each, a water-melon for ₹ 23.50. The cost of empty basket is ₹ 5.50. How many such baskets can be purchased for ₹ 6675?
- A cloth shrinks 0.8% when washed. What fraction is this?



ANSWERS

Exercise 3.1

- (i) $\frac{4}{5}$ (ii) $\frac{9}{4}$ (iii) $\frac{35}{2}$ (iv) $\frac{3}{80}$ (v) $\frac{131}{250}$
- (i) 2.9 (ii) 15.125 (iii) 234.04 (iv) 18.304
- (i) 37.86 (ii) 0.054 (iii) 22.475 (iv) 38.674
- (i) 3.852 (ii) 23.55 (iii) 63.52 (iv) 5.715
- (i) 59.79 (ii) 29.41 (iii) 22.271 (iv) 6.5693
- (i) 0.22, 4.6, 20.05, 24.13
(ii) 0.52, 2.05, 3.5, 36.28

Exercise 3.2

- (i) 26 (ii) 347 (iii) 1286 (iv) 2340.5
(v) 211 (vi) 35240 (vii) 357 (viii) 4806 (ix) 7
(x) 8 (xi) 90 (xii) 60
- (i) 3.6 (ii) 31.15 (iii) 21.6 (iv) 176.4 (v) 0.4
(vi) 1404.24 (vii) 1.65
- (i) 1.24 (ii) 2.17 (iii) 121 (iv) 0.009 (v) 4.462
(vi) 41.1025 (vii) 1.716 (viii) 0.0315
(ix) 60.6404 (x) 1.0101 (xi) 440.044
- 33.64 sq. cm 5. ₹ 341.55 6. ₹ 713.25
7. ₹ 952 8. 131.13 m²
- (i) 0.01728 (ii) 12 (iii) 36.2637 (iv) 0.000036

Exercise 3.3

- (i) 0.25 (ii) 0.025 (iii) 0.0025 (iv) 5.56
(v) 0.0556 (vi) 0.369 (vii) 0.056 (viii) 0.326
(ix) 0.0379 (x) 2.337

- (i) 51 (ii) 9 (iii) 3.4 (iv) 19 (v) 900 (vi) 5 (vii) 27
(viii) 2.1 (ix) 10
- (i) 3.245 (ii) 90 (iii) 4.42 (iv) 0.31 (v) 0.7
(vi) 0.535 (vii) 2.055 (viii) 0.025
- (i) 1.5 (ii) 0.96 (iii) 67.5 (iv) 0.25 (v) 0.375
(vi) 0.048 (vii) 0.425 (viii) 0.12 (ix) 0.104
- ₹ 9.60 6. 63 7. 6.23 8. ₹ 21.35 9. ₹ 42.60
10. ₹ 22

Multiple Choice Questions

- (c) 2. (d) 3. (a) 4. (c) 5. (d)
6. (b) 7. (a) 8. (b) 9. (b) 10. (c)

Mental Maths Corner

- 36.696 2. 100 3. 1.7 4. 0.005 5. 2.5
6. 3.42 7. 0.01 8. right 9. 85 kg 10. 1.3

Review Exercise

- 7.04, 7.06, 7.4, 7.46, 7.6, 7.64
- (i) 0.8 (ii) 6.528 (iii) 1.56 (iv) 3.625
- 29.504 4. 3.3796
- (i) 32500 (ii) 3.555 (iii) 0.026 (iv) 3.924
- (i) 19.24 (ii) 0.3408 (iii) 0.00416 (iv) 0.00156
- (i) 17 (ii) 34.5 (iii) 75 (iv) 0.76
- 0.48 9. ₹ 72.5 10. 16.4025 sq. m
11. 67.575 sq. m 12. ₹ 360.25 13. 75

HOTS Questions

- 6, hexagon 2. 75 3. $\frac{1}{125}$